

COMPSCI 1DM3 - Winter 2026

Discrete Mathematics for Computer Science

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Last Lecture:

- Propositional function $P(x)$: reduces to a proposition for a fixed x
- Quantifiers:
 - ▶ Universal: $\forall x P(x)$ is true if $P(x)$ is true for all x in the domain
 - ▶ Existential: $\exists x P(x)$ is true if there exists (at least) an x in the domain for which $P(x)$ is true
- Negation:
 - ▶ $\neg \forall x P(x) \equiv \exists x \neg P(x)$
 - ▶ $\neg \exists x P(x) \equiv \forall x \neg P(x)$
- Nested quantifiers:
 - ▶ Example $\forall x \forall y P(x, y), \exists x \exists y P(x, y), \forall x \exists y P(x, y)$
 - ▶ $\exists x \forall y P(x, y) \rightarrow \forall y \exists x P(x, y)$, but not the other way!

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- Argument: Given a sequence of propositions $p_1, p_2, \dots, p_n$ and q , we wish to infer q (conclusion) from $p_1, p_2, \dots, p_n$ (premises):
If $p_1, p_2, \dots, p_n$ are all True, then q is true.
Equivalently: $(p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow q$ is a tautology.

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- We denote this inference as

$$\begin{array}{c} p_1 \\ p_2 \\ \vdots \\ p_n \\ \hline \text{therefore } \therefore q \end{array}$$

Rules of inference

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Why? Because $(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$ is a tautology.

More rules of inference

- Hypothetical syllogism:

$$p \rightarrow q$$

$$q \rightarrow r$$

$$\therefore p \rightarrow r$$

More rules of inference

p : You submit the A.
 q : You pass the course
 r : you graduate.

- Hypothetical syllogism:

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$$

if you submit the A, you pass the course
if you pass the course, you graduate.

Why? Because $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (\underline{p} \rightarrow \underline{r})$ is a tautology.

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- Disjunctive syllogism:

$$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$$

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- Disjunctive syllogism:

$$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$$

*p: the server is down
q: the cable is unplugged.*

$p \vee q$: the server is down or the cable is unplugged.

$\neg p$: the server is not down.

Why? Because $((p \vee q) \wedge \neg p) \rightarrow q$ is a tautology.

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$$\neg(p \wedge q) \vee q$$

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Why? Because $(p \wedge q) \rightarrow p \wedge q$ is a tautology (trivially!).

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p: printer is out of ink
q: paper tray is empty.
r: the printer is broken.

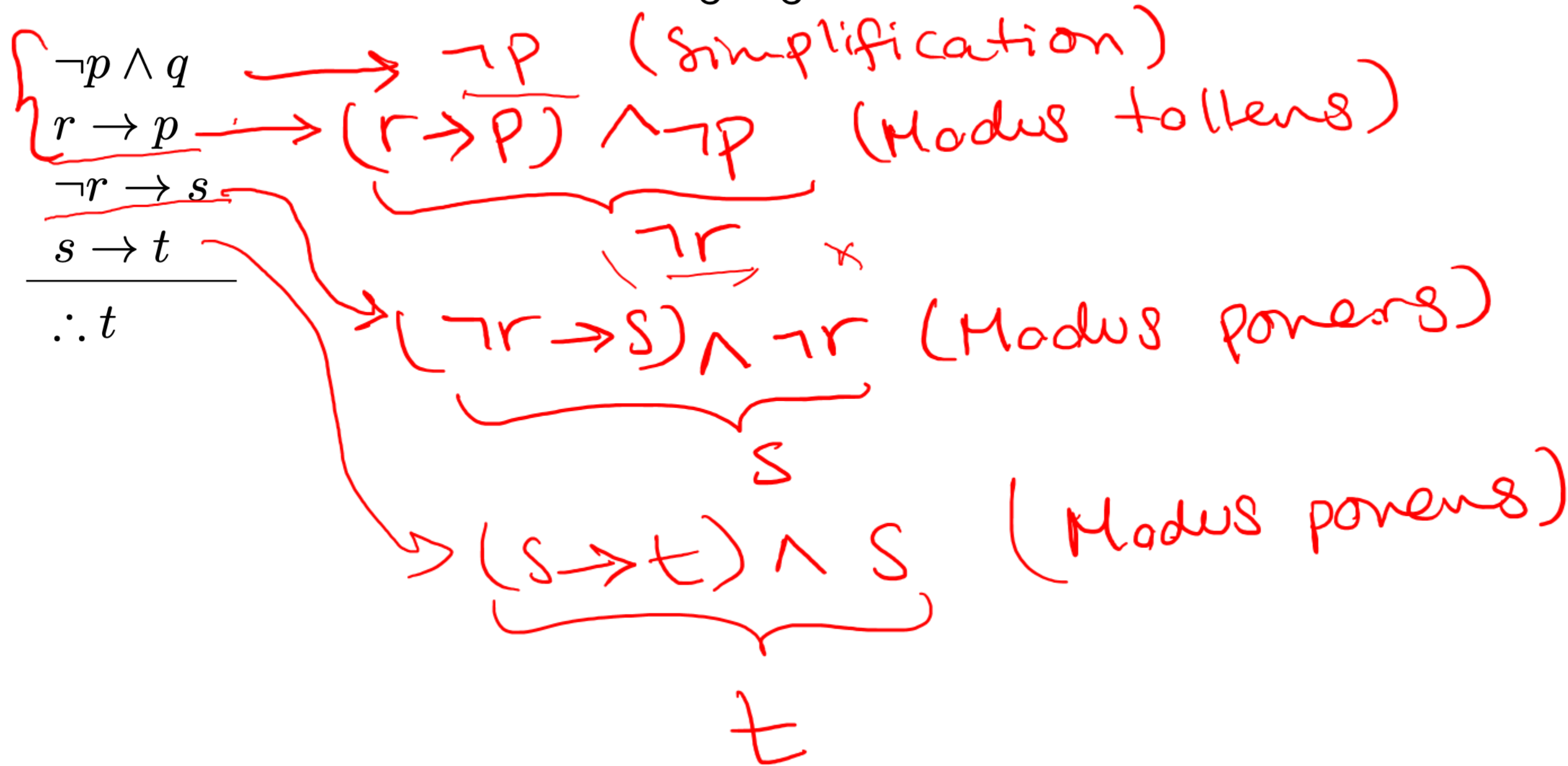
Why? Because $((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$ is a tautology.

Rules of inference: Summary

- Modus ponens: $p \rightarrow q$ and p therefore q
- Modus tollens: $p \rightarrow q$ and $\neg q$ therefore $\neg p$ $(p \rightarrow q) \wedge \neg q$
- Hypothetical syllogism: $p \rightarrow q$ and $q \rightarrow r$ therefore $p \rightarrow r$
- Disjunctive syllogism: $p \vee q$ and $\neg p$ therefore q
- Addition: p therefore $p \vee q$
- Simplification: $p \wedge q$ therefore q
- Conjunction: p and q therefore $p \wedge q$ (trivially!)
- Resolution: $p \vee q$ and $\neg p \vee r$ therefore $q \vee r$

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Determine whether the following argument is valid:



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$$\frac{(p \wedge q) \vee r \quad r \rightarrow s}{\therefore p \vee s}$$

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- Modus tollens: $p \rightarrow q$ and $\neg q$ therefore $\neg p$
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- Resolution: $p \vee q$ and $\neg p \vee r$ therefore $q \vee r$

Examples

Determine whether the following argument is valid:

$$p \rightarrow r$$

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Rules of inference for quantifiers:

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- What do we mean by “mathematical statement”:
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 - ▶ Lemma: a result that is needed to prove a theorem
 - ▶ Corollary: a result that directly follows from a theorem

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- Proof techniques:
 - ▶ Direct proof
 - ▶ Proof by contraposition
 - ▶ Proof by contradiction
 - ▶ Proof by cases

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Example. If n is an odd integer, then n^2 is odd.